

AADHUNIK SHARMA

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Summary

Quantitative researcher with over 6 years of experience specializing in algorithmic trading and financial market analysis. Proficient in Python, C#, and C++. Experienced across all market segments, enabling the development and implementation of effective quantitative strategies.

Education

World Quant University

2019 - 2021

M.Sc. Financial Engineering

USA

Sant Longowal Institute of Engineering and Technology

2015 - 2019

B.E Computer Science

India

Skills

Technologies: Data Analytics, Backtesting, Machine learning

Languages: Python, SQL, C#, C++

Databases: MySQL, SQLite, InfluxDB, MongoDB

Visualization : Plotly, Dash, Bokeh, Matplotlib

Libraries : Pandas, Numpy, Numba, Scikit-learn , Keras, Talib

Experience

Quan Algorithms - Senior Python Developer

May 2020 – present

Performed large-scale data analytics on financial microstructure data across **Forex, Futures, Options, and Cryptocurrency markets** using **Python** and **SQL**. Built self-service analytic solutions for data extraction, transformation, and analysis. Developed a high-performance backtesting system optimized with **Numba** and **multiprocessing** to handle large-scale datasets efficiently.

Engineered advanced **quantitative trading systems**, implementing a **VIX-based hedging strategy** using the **Hidden Markov Model (HHMM)** to dynamically adjust risk exposure. Designed an **option beta decay strategy** to exploit theta decay and hedge against volatility shifts.

Developed **anomaly detection algorithms** leveraging **machine learning** techniques to identify irregular trading patterns and market inefficiencies. Integrated predictive models to enhance trading performance across diverse markets.

Built **interactive visual dashboards** using **Plotly, Bokeh, and InfluxDB** to present actionable insights and enable real-time decision-making.

Curl Analytics - Intern Algo Trading Strategist

Nov 2019 – May 2020

Analyzed and backtested large market data sets to develop beta-neutral index options strategies exploiting theta decay. Performed data munging and transformation for clean, structured data processing. The role involved backtesting strategies, refining risk management techniques, and optimizing performance through iterative model adjustments. Strategies included calendar spreads between futures and selling option straddles with time-based factors to capitalize on volatility and time decay dynamics.

Projects

Beta-Neutral Index Option Strategy | *Python, SQL, Numba*

- Developed several beta-neutral intraday index option strategies with a Sharpe ratio of 2.5+ using advanced backtesting, optimization, and machine learning.
- Developed dashboards in Plotly to visualize market data and regime anomalies.
- Employed genetic algorithms for variable optimization and machine learning model XGBoost.
- Optimized backtesting framework for speed and scalability with Numba and multiprocessing in Python.

Crude Oil Inventory Trading Model | *Python*

- Developed a predictive trading model for crude oil inventory movements using historical weekly inventory and market data.
- Engineered features from supply, demand, and macroeconomic indicators to forecast inventory-driven price direction.

Cryptocurrency Arbitrage Bot | *Python, SQL*

- Developed a cryptocurrency arbitrage bot executing across two major exchanges to capitalize on price discrepancies.
- Designed the system for minimal market exposure, automated trade execution, and optimized profit opportunities.
- Implemented continuous monitoring and rebalancing to mitigate risk.

TelegramFXCopier Software Development | *C#, SQL*

- Led a team of 4 developers to enhance and maintain **TelegramFXCopier**, a cutting-edge product for automated trade copying.
- Developed new modules and implemented critical fixes to improve the performance, scalability, and reliability of the system.

Achievements

- * Achieved **1 Silver and 6 Bronze** medals in competitive coding on HackerRank, alongside a **Golden Algorithmic Badge**.
- * Ranked in the **top 10% globally** in competitive programming hackathons on HackerRank.